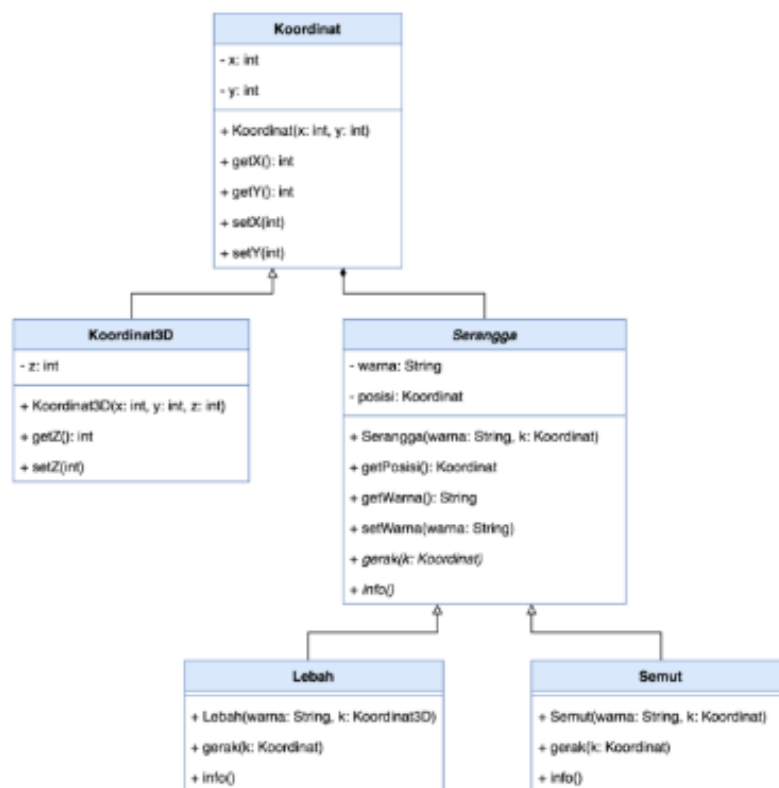


Tugas Pendahuluan Modul 9
PEMROGRAMAN BERORIENTASI OBJEK – Ganjil 2025/2026
“Polimorfisme, Kelas Statik, & Collection”

Nama : Syahril Arfian Almazril
Nim : 103032300013
Kelas : IT-47-04

Soal Tugas Pendahuluan :

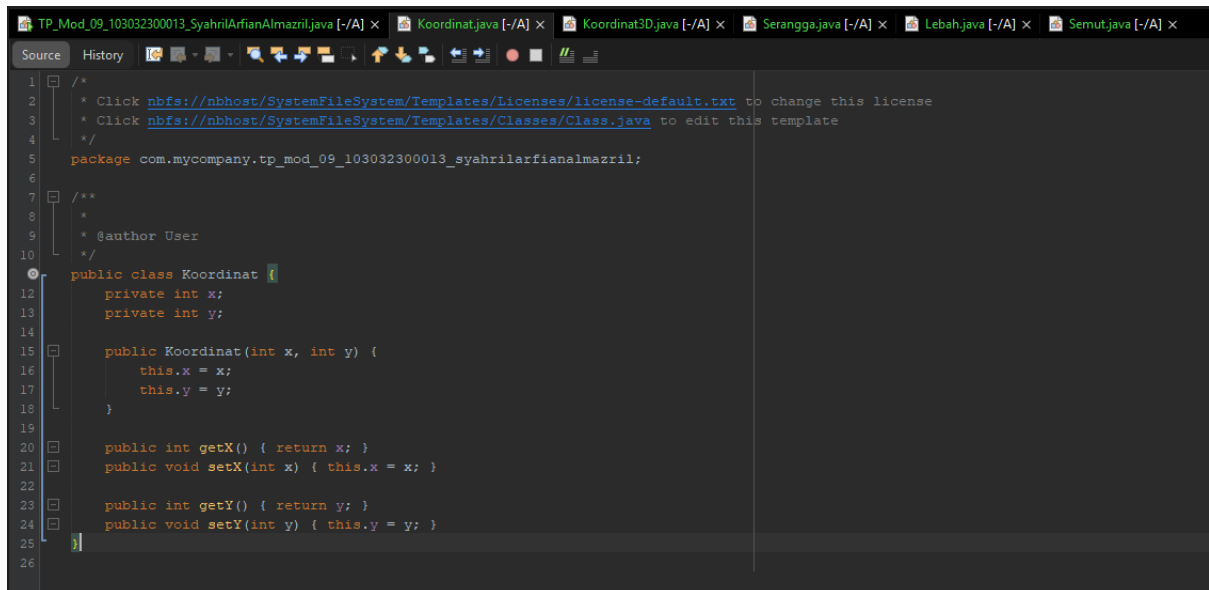


Berikut penjelasan atau contoh beberapa atribut dan method yang ada di class diagram di atas:

1. Buatlah implementasi kelas dari class diagram di atas, letakkan semuanya di package **"TPO8"**.
2. Implementasi **konstruktor** pada kelas **Lebah** dan **Semut** dengan parameter warna dan Koordinat (Semut) / Koordinat3D (Lebah).
4. Implementasi method **info()** untuk menampilkan warna dan titik posisi dari serangga.
5. Selanjutnya pada class **"main.java"** (yang berisi main method saat project dibuat), lakukan uji coba dengan scenario membuat objek-objek untuk kelas Serangga sehingga hasilnya adalah sebagai berikut:

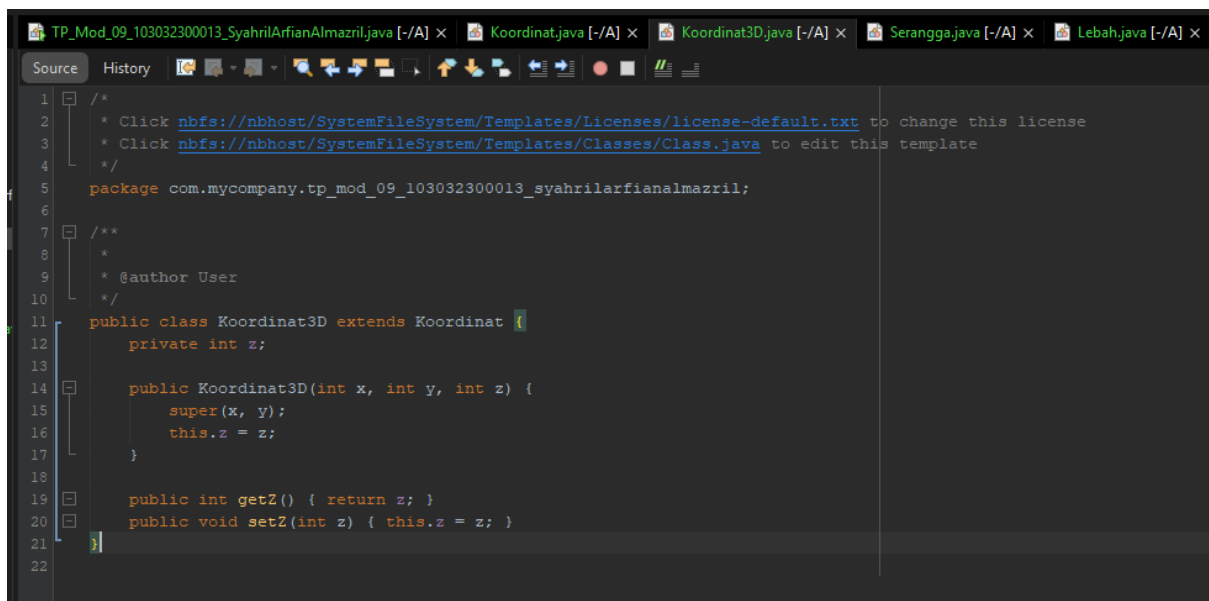
Tugas Pendahuluan Modul 9
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“Polimorfisme, Kelas Statik, & Collection”

Koordinat.java



```
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4   */
5   package com.mycompany.tp_mod_09_103032300013_syahrilarfianalmazril;
6
7   /**
8    *
9    * @author User
10   */
11  public class Koordinat {
12      private int x;
13      private int y;
14
15      public Koordinat(int x, int y) {
16          this.x = x;
17          this.y = y;
18      }
19
20      public int getX() { return x; }
21      public void setX(int x) { this.x = x; }
22
23      public int getY() { return y; }
24      public void setY(int y) { this.y = y; }
25  }
```

Koordinat3D



```
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4   */
5   package com.mycompany.tp_mod_09_103032300013_syahrilarfianalmazril;
6
7   /**
8    *
9    * @author User
10   */
11  public class Koordinat3D extends Koordinat {
12      private int z;
13
14      public Koordinat3D(int x, int y, int z) {
15          super(x, y);
16          this.z = z;
17      }
18
19      public int getZ() { return z; }
20      public void setZ(int z) { this.z = z; }
21  }
```

Tugas Pendahuluan Modul 9
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Serangga.java

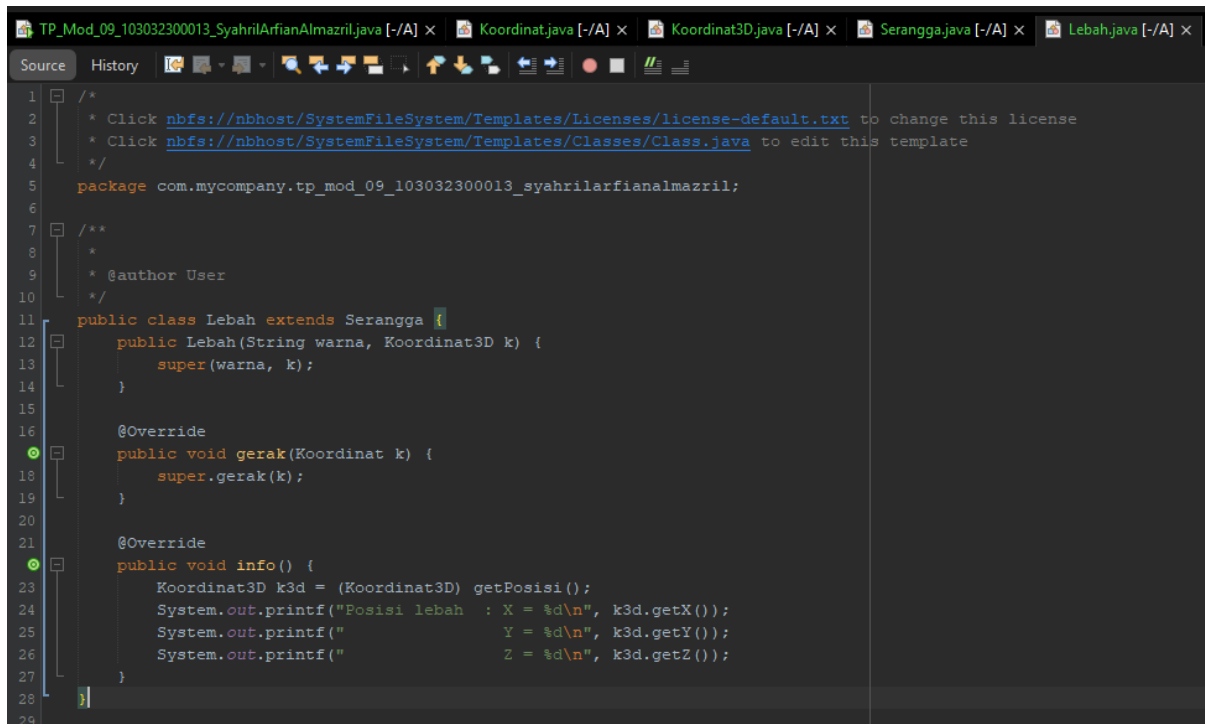
```
TP_Mod_09_103032300013_SyahrilArfianAlmazril.java [-/A] × Koordinat.java [-/A] × Koordinat3D.java [-/A] × Serangga.java [-/A] × Lebah.java [-/A] ×
Source History
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4   */
5   package com.mycompany.tp_mod_09_103032300013_syahrilarfianalmazril;
6
7   /**
8    *
9    * @author User
10   */
11   public class Serangga {
12       private String warna;
13       private Koordinat posisi;
14
15       public Serangga(String warna, Koordinat posisi) {
16           this.warna = warna;
17           this.posisi = posisi;
18       }
19
20       public Koordinat getPosisi() { return posisi; }
21       public String getWarna() { return warna; }
22
23       public void setWarna(String warna) { this.warna = warna; }
24
25       public void gerak(Koordinat k) { this.posisi = k; }
26
27       public void info() {
28           System.out.printf("Warna serangga : %s\n", warna);
29           System.out.printf("Posisi serangga : X = %d\n", posisi.getX());
30           System.out.printf("                Y = %d\n", posisi.getY());
31       }
32   }
33
```

Semut.java

```
TP_Mod_09_103032300013_SyahrilArfianAlmazril.java [-/A] × Koordinat.java [-/A] × Koordinat3D.java [-/A] × Serangga.java [-/A] × Lebah.java [-/A] ×
Source History
1  /*
2   * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3   * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4   */
5   package com.mycompany.tp_mod_09_103032300013_syahrilarfianalmazril;
6
7   /**
8    *
9    * @author User
10   */
11   public class Semut extends Serangga {
12       public Semut(String warna, Koordinat k) {
13           super(warna, k);
14       }
15
16       @Override
17       public void gerak(Koordinat k) {
18           super.gerak(k);
19       }
20
21       @Override
22       public void info() {
23           System.out.printf("Posisi semut : X = %d\n", getPosisi().getX());
24           System.out.printf("                Y = %d\n", getPosisi().getY());
25       }
26   }
27
```

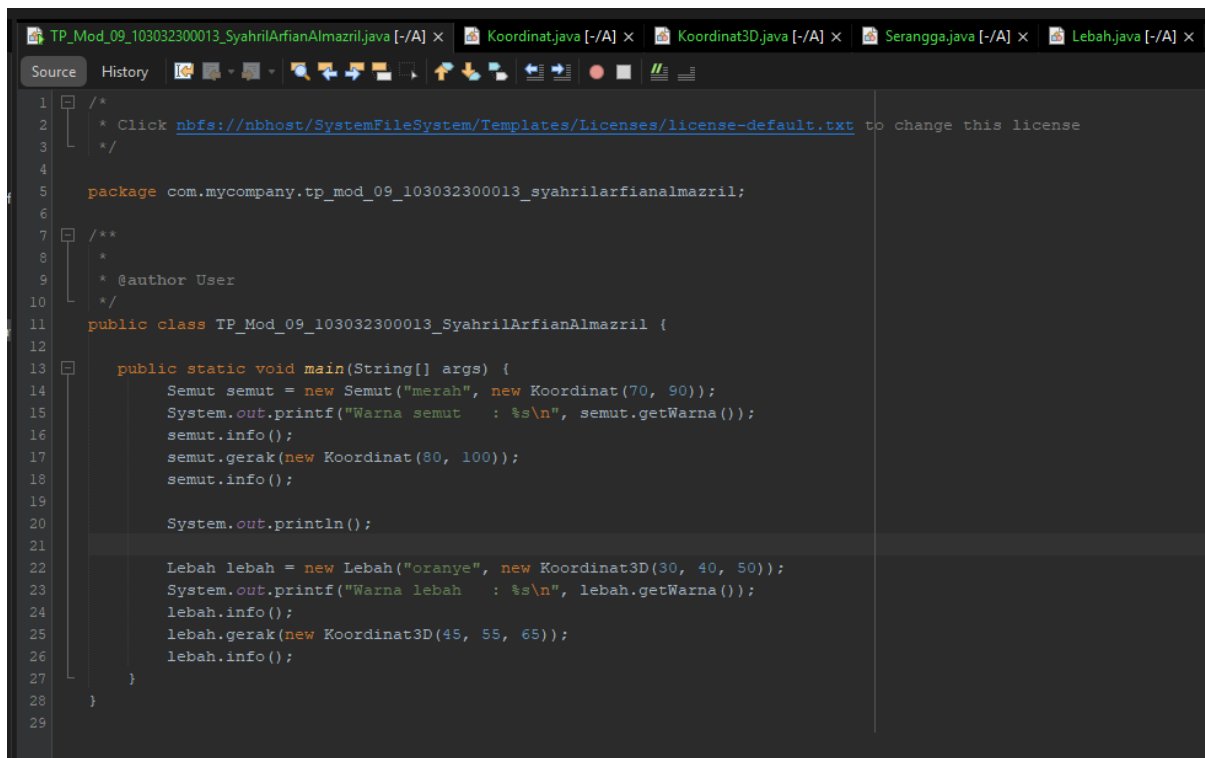
Tugas Pendahuluan Modul 9
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“Polimorfisme, Kelas Statik, & Collection”

Lebah.java



```
1  /*
2  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3  * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4  */
5  package com.mycompany.tp_mod_09_103032300013_syahrilarfianalmazril;
6
7  /**
8   *
9   * @author User
10  */
11  public class Lebah extends Serangga {
12      public Lebah(String warna, Koordinat3D k) {
13          super(warna, k);
14      }
15
16      @Override
17      public void gerak(Koordinat k) {
18          super.gerak(k);
19      }
20
21      @Override
22      public void info() {
23          Koordinat3D k3d = (Koordinat3D) getPosisi();
24          System.out.printf("Posisi lebah : X = %d\n", k3d.getX());
25          System.out.printf("                Y = %d\n", k3d.getY());
26          System.out.printf("                Z = %d\n", k3d.getZ());
27      }
28  }
29
```

Main.java



```
1  /*
2  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3  * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4  */
5  package com.mycompany.tp_mod_09_103032300013_syahrilarfianalmazril;
6
7  /**
8   *
9   * @author User
10  */
11  public class TP_Mod_09_103032300013_Syahrilarfianalmazril {
12
13      public static void main(String[] args) {
14          Semut semut = new Semut("merah", new Koordinat(70, 90));
15          System.out.printf("Warna semut : %s\n", semut.getWarna());
16          semut.info();
17          semut.gerak(new Koordinat(80, 100));
18          semut.info();
19
20          System.out.println();
21
22          Lebah lebah = new Lebah("oranye", new Koordinat3D(30, 40, 50));
23          System.out.printf("Warna lebah : %s\n", lebah.getWarna());
24          lebah.info();
25          lebah.gerak(new Koordinat3D(45, 55, 65));
26          lebah.info();
27      }
28  }
29
```

Tugas Pendahuluan Modul 9
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Output :

```
Output x
Run (TP_Mod_09_103032300013_SyahriArfianAlmazri) x Documents - C:\Users\User\OneDrive\Documents x
cd C:\Users\User\OneDrive\Documents\NetBeansProjects\TP_Mod_09_103032300013_SyahriArfianAlmazri; "JAVA_HOME=C:\Program Files\Apache NetBeans\jdk" cmd /c "%C:\Program Files\Apache NetBeans\jdk\bin\java.exe" %*
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::staticFieldBase has been called by com.google.inject.internal.aop.HiddenClassDefiner (file:/C:/Program%20Files/Apache%20NetBeans/java/maven/lib/guice-5.1.0-Classes.jar)
WARNING: Please consider reporting this to the maintainers of class com.google.inject.internal.aop.HiddenClassDefiner
WARNING: sun.misc.Unsafe::staticFieldBase will be removed in a future release
Scanning for projects...

-----< com.mycompany:TP_Mod_09_103032300013_SyahriArfianAlmazri >-----
Building TP_Mod_09_103032300013_SyahriArfianAlmazri 1.0-SNAPSHOT
from pom.xml
-----[ jar ]-----

--- resources:3.1.1:resources (default-resources) @ TP_Mod_09_103032300013_SyahriArfianAlmazri ---
skip non existing resourceDirectory C:\Users\User\OneDrive\Documents\NetBeansProjects\TP_Mod_09_103032300013_SyahriArfianAlmazri\src\main\resources

--- compiler:3.13.0:compile (default-compile) @ TP_Mod_09_103032300013_SyahriArfianAlmazri ---
Recompiling the module because of changed source code.
Compiling 6 source files with javac [debug release 24] to target\classes

--- exec:3.1.0:exec (default-cli) @ TP_Mod_09_103032300013_SyahriArfianAlmazri ---
Warna semut : merah
Posisi semut : X = 70
               Y = 60
Posisi semut : X = 80
               Y = 100

Warna lebah : oranye
Posisi lebah : X = 30
               Y = 40
               Z = 50
Posisi lebah : X = 45
               Y = 55
               Z = 65

BUILD SUCCESS

Total time: 2.357 s
Finished at: 2025-11-08T23:46:33+07:00
```